SQL Practice Exercise 4: Joins, Union, Filtering and Aggregates

Query 1: Retrieve all employees and their assigned projects using an Inner Join

```
Select t1.employee_id, first_name, last_name, dept, salary
        t2. project_id As Project_ID
        t2. project_name As project_name, t2.Budget, status
From 'Workspace. default. real. table_1_emp_Join_union_1 t1'
Inner Join 'Workspace. default. real. projectID_employee_Joins t2'
On t1.Employee_ID = t2. employee_id;
```

mp_ID	First_Name	last_Name	Dept	Salary	Pro ID	ProName	Bud	Status
1	John	Doe	IT	70000	101	AI Dev	100000	Complete
2	Alice	Smith	HR	60000	102	Empl Tr	50000	On-Going
3	Bob	Johnson	finance	75000	103	CA	75000	Pending
4	David	Brown	IT	72000	104	FA	90000	On-Going
5	Emma	Wilson	Sales	65000	105	ME	65000	Completed
6	Michael	Clark	finance	80000	106	RM	80000	Pending

Query 2: Retrieve all employees & their assigned projects, employees who have no projects using Left Join

```
Select emp_id, first_name, last_name, department, salary, project_ID As Project_ID
project_name As project_name
From 'Workspace. default. real. table_1_emp_Join_union.'
Left Join 'Workspace. default. real. projectId_employee_joins 2'
On t1. employee_id = t2. employee_id;
```

Query 3: Retrieve all projects & their assigned employees, including projects that have no employees using Right Join

```
Select emp-id, first-name, last-name, dep, salary, project-id As project-id
project-name As project-name, budget, status
From 'Workspace.default.real-table-1-employee-Joins' t1
Right Join 'Workspace.default.real-projectId-emp-Joins' t2
On t1.employee-id = t2.employee-id;
```

Query 4: Retrieve all employees and projects, including those without a match in either table using full Outter Join

```
Select emp-id, first-name, last-name, dept, salary, project-ID As project-ID
project-name As project-Name, budget, status
From 'Workspace.default.real-table-1-employee-Joins t1'
Full Outter Join 'Workspace.default.real-projectID-employee-Join
On t1.Employee-ID = t2.Employee-ID;
```

Query 5: Retrieve a list of all unique cities where employees are located.

```
Select Distinct city As location
From 'Workspace.default.real-table-1-employee'
Union
Select Distinct Status As location
From 'Workspace.default.real-projectID-employees';
```

City	Status
London	On-Going
New York	Complete
Los Angeles	On-Going
Sydney	Complete
Toronto	Pending

Query 6: Retrieve a list of all cities where employees are located & projects statuses, allowing duplicates

```
Select city As location
From 'Workspace.default.real_table_1_employee_Joins'
Union All
Select status As Location
From 'Workspace.default.real_project_ID_employee_Joins';
```

City	Statuses
New York	Completed
Los Angeles	On-Going
Toronto	Pending
London	On-Going
Sydney	Completed
New York	Pending

Query 7: Retrieve employees who earn more than 70000

```
Select Emp_ID, First_name, last_name, department, salary
From 'Workspace.default.real_table_1_employee_Joins'
Where Salary > 70000;
```

Emp_ID	First Name	Last Name	Dept	Salary
3	Bob	Johnson	Finance	75000
4	David	Brown	IT	72000
6	Micheal	Clark	Finance	80000

Query 8: Retrieve employees working in either IT or Finance department.

```
Select emp-id, first-name, last-name, department, Salary
From 'Workspace. default. real-table-employee-Joins'
Where department IN ('IT', Finance');
```

Emp-ID	First Name	last Name	Dep	Salary
1	John	Doe	IT	70000
3	Bob	Johnson	Finance	75000
4	David	Brown	IT	72000
6	Michael	Clark	Finance	80000

Query 9: Retrieve projects that are not yet completed.

```
Select project-Id, project-name, budget, Status
From 'Workspace. default-real-projectID-employee-Joins'
Where Status <> 'Completed';
```

Project-ID	Project Name	Budget	Status
102	Employee Training	50000	On-Going
103	Cybersecurity Audit	75000	Pending
104	Financial Analysis	90000	On-Going
106	Risk Management	80000	Pending

Query 10: Retrieve projects that have a budget greater than 70,000 and are not completed

```
Select project_ID, Project_name, Budget, Status
From 'Workspace.default.real_projectId_employee_Joins'
Where Budget > 70000
AND Status <> 'Completed';
```

Project_ID	Project_Name	Budget	Status
103	Cybersecurity Audit	75000	Pending
104	Financial Analysis	90000	On-Going
106	Risk Management	80000	Pending

Query 11: Retrieve employees from New York or Toronto, ordered by sal in desc order

```
Select Emp_ID, first_name, last_name, dept, salary, city
From 'Workspace.default.real_table_1_employee_Joins'
Where City IN ('New York', 'Toronto')
Order By Salary Desc
```

Emp_ID	First_Name	Last_Name	Dept	Sal	City
6	Michael	Clark	Finance	80000	New York
3	Bob	Johnson	Finance	75000	Toronto
1	John	Doe	IT	70000	New York

Query 12: Retrieve the top 3 highest-paid employees

```
Select Emp-ID, first-name, last-name, Dep, Salary
From 'Workspace.default.real-table-1-employee-Joins'
Order By Salary Desc
Limit 3;
```

Emp-ID	First Name	Last Name	Dept	Salary
6	Michael	Clark	Finance	80000
3	Bob	Johnson	Finance	75000
4	David	Brown	IT	72000

Query 13: Find the total salary per department, sorted in descending order

*ANSWER

Department	Total Salary
Finance	155000
IT	142000
Sales	65000
HR	60000

```
Select Department, Sum (Salary) As total Salary
From 'Workspace.default.real-table-1-employee-Join'
Group By Department
Order By Total Salary Desc;
```

Query 14: Find the average salary per city, but only include cities where the average salary is greater than 65000

```
Select City AVG (salary) As Average Salary
From 'Workspace.default.real_table_1_employee_joins'
Group By City
Having AVG (salary) > 65000;
```

City	Average Salary
New York	75000
Toronto	75000
London	72000

Query 15: Count the number of employees per department, including only departments, with more than 1 employee

```
Select Department, Count (Employee_ID) As Employee Count
From 'Workspace.default.real_table_1_employee_Joins'
GroupBy Department
Having Count (Employee_ID) > 1;
```

Department	Employee Count
IT	2
Finance	2

Query 16: Retrieve the number of projects per status, but only include statuses with at least 2 projects.

```
Select Status, Count (project_ID) As project Count
From 'Workspace. default. real-projectID_employee_Joins'
Group By Status
Having Count (project_ID) >= 2;
```

Status	Project Count
Complete	2
On-Going	2
Pending	2

Query 17: Retrieve the total project budget per employee, but only for employees who are managing projects worth more than 150,000

```
Select emp_ID, first_name, last_name, Sum (Budget) As total_project_Budget
From 'Workspace. default. real-table-1-employee_Joins'
Join 'Workspace. default. real-projectID_employee_Joins'
On t1.employee_ID = t2. Employee_ID
Group By employee_ID, First_names, last_names
Having Sum (Budget) > 150000
```